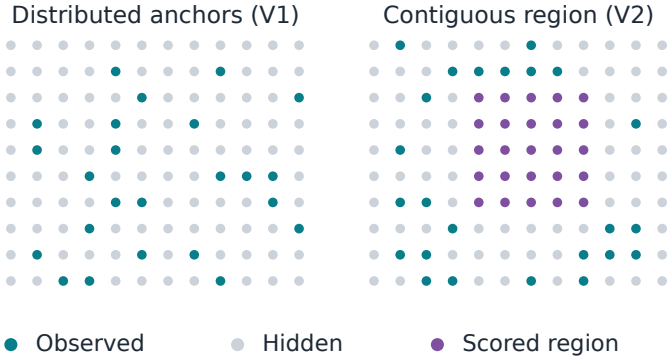
a Cohort roles and observation geometry



Development: HEST, 4 patients

2 train / 1 validation / 1 evaluation

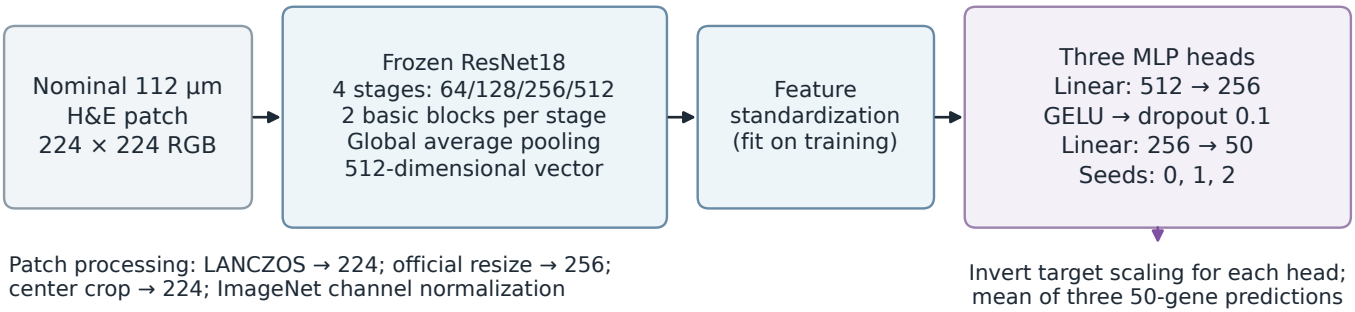
Transportability: HER2ST, 8 patients

Reused for calibration development

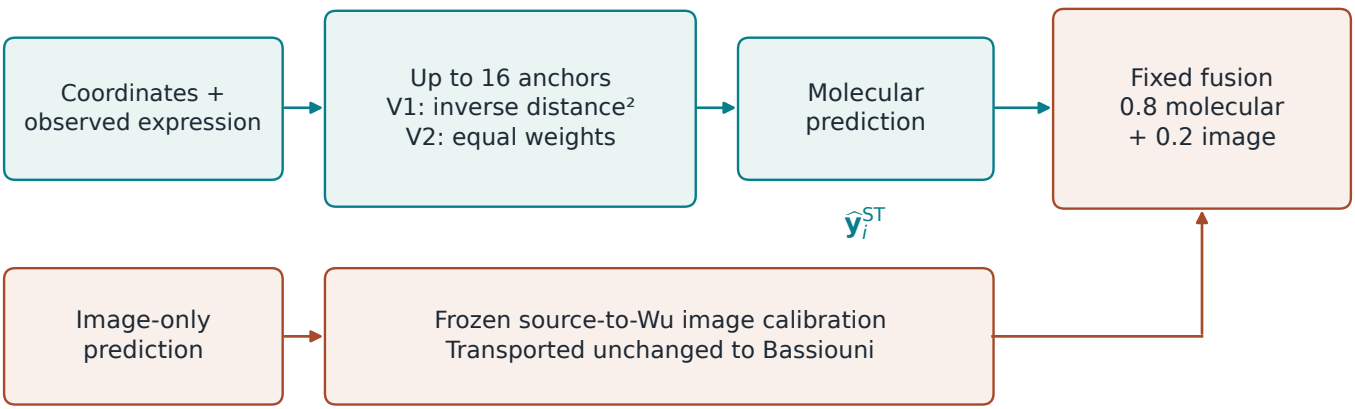
Fresh evaluation: Wu 3 + Bassiouni 22

Three separate Wu patients supply calibration

b Frozen encoder and three trained regression heads



c Molecular interpolation and fixed fusion



Score hidden locations only: image, molecular and fusion predictions

V1 scores all hidden locations; V2 scores the designated contiguous region.